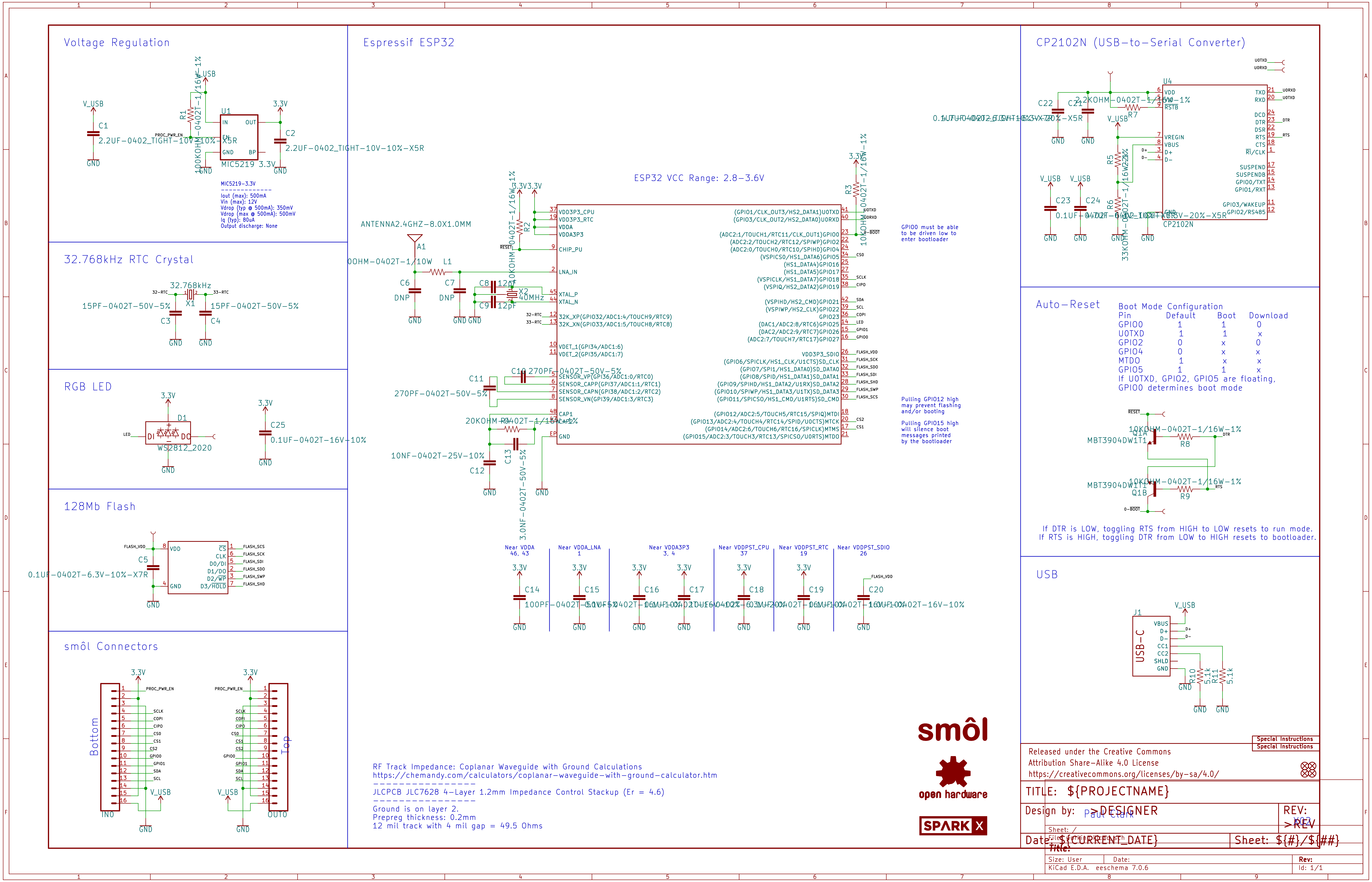

Voltage Regulation
V_USB
C1
2.2UF-0402_TIGHT-10V-10%-X5R
GND
R1
100KOHM-0402T-1/16W-1%
PROC_PWR_EN
U1
IN
OUT
EN
GND
BP
MIC5219 3.3V
3.3V
C2
2.2UF-0402_TIGHT-10V-10%-X5R
MIC5219-3.3V
Iout (max): 500mA
Vin (max): 12V
Vdrop (typ @ 500mA): 350mV
Vdrop (max @ 500mA): 500mV
Iq (typ): 80uA
Output discharge: None
32.768kHz RTC Crystal
32.768kHz
X1
15PF-0402T-50V-5%
C3
C4
RGB LED
3.3V
D1
DI
DO
WS2812_2020
C25
0.1UF-0402T-16V-10%
128Mb Flash
FLASH_VDD
VDD
C5
0.1UF-0402T-6.3V-10%-X7R
CS
CLK
DO/DI
D1/DO
D2/WP
D3/HOLD
FLASH_SCS
FLASH_SCK
FLASH_SDI
FLASH_SDO
FLASH_SWP
FLASH_SHD
smôl Connectors
Bottom
Top
IN0
OUT0
PROC_PWR_EN
SCLK
COPI
CIPO
CS0
CS1
CS2
GPIO0
GPIO1
SDA
SCL
V_USB
Espressif ESP32
ESP32 VCC Range: 2.8-3.6V
ANTENNA2.4GHZ-8.0X1.0MM
A1
L1
C6
C7
DNP
C8
C9
12pF
X2
40MHz
R2
10KOHM-0402T-1/16W-1%
RESET
VDD3P3_CPU
VDD3P3_RTC
VDDA
VDDA3P3
CHIP_PU
LNA_IN
XTAL_P
XTAL_N
32K_XP(GPIO32/ADC1:4/TOUCH9/RTC9)
32K_XN(GPIO33/ADC1:5/TOUCH8/RTC8)
VDET_1(GPI34/ADC1:6)
VDET_2(GPI35/ADC1:7)
SENSOR_VP(GPI36/ADC1:0/RTC0)
SENSOR_CAPP(GPI37/ADC1:1/RTC1)
SENSOR_CAPN(GPI38/ADC1:2/RTC2)
SENSOR_VN(GPI39/ADC1:3/RTC3)
CAP1
CAP2
GND
C10 270PF-0402T-50V-5%
C11
270PF-0402T-50V-5%
20KOHM-0402T-1/16W-1%
R4
10NF-0402T-25V-10%
C12
C13
3.0NF-0402T-50V-5%
(GPIO1/CLK_OUT3/HS2_DATA1)U0TXD
(GPIO3/CLK_OUT2/HS2_DATA0)U0RXD
(ADC2:1/TOUCH1/RTC11/CLK_OUT1)GPIO0
(ADC2:2/TOUCH2/RTC12/SPIWP)GPIO2
(ADC2:0/TOUCH0/RTC10/SPIHD)GPIO4
(VSPICS0/HS1_DATA6)GPIO5
(HS1_DATA4)GPIO16
(HS1_DATA5)GPIO17
(VSPICLK/HS1_DATA7)GPIO18
(VSPIQ/HS2_DATA2)GPIO19
(VSPIHD/HS2_CMD)GPIO21
(VSPIWP/HS2_CLK)GPIO22
GPIO23
(DAC1/ADC2:8/RTC6)GPIO25
(DAC2/ADC2:9/RTC7)GPIO26
(ADC2:7/TOUCH7/RTC17)GPIO27
VDD3P3_SDIO
(GPIO6/SPICLK/HS1_CLK/U1CTS)SD_CLK
(GPIO7/SPI1/HS1_DATA0)SD_DATA0
(GPIO8/SPID/HS1_DATA1)SD_DATA1
(GPIO9/SPIHD/HS1_DATA2/U1RXD)SD_DATA2
(GPIO10/SPIWP/HS1_DATA3/U1TXD)SD_DATA3
(GPIO11/SPICS0/HS1_CMD/U1RTS)SD_CMD
(GPIO12/ADC2:5/TOUCH5/RTC15/SPIQ)MTDI
(GPIO13/ADC2:4/TOUCH4/RTC14/SPID/U0CTS)MTCK
(GPIO14/ADC2:6/TOUCH6/RTC16/SPICLK)MTMS
(GPIO15/ADC2:3/TOUCH3/RTC13/SPICS0/U0RTS)MTDO
R3
10KOHM-0402T-1/16W-1%
GPIO0 must be able to be driven low to enter bootloader
Pulling GPIO12 high may prevent flashing and/or booting
Pulling GPIO15 high will silence boot messages printed by the bootloader
Near VDDA 46, 43
Near VDDA_LNA 1
Near VDDA3P3 3, 4
Near VDDPST_CPU 37
Near VDDPST_RTC 19
Near VDDPST_SDIO 26
C14
C15
C16
C17
C18
C19
C20
100PF-0402T-50V-5%
0.1UF-0402T-16V-10%
RF Track Impedance: Coplanar Waveguide with Ground Calculations
https://chemandy.com/calculators/coplanar-waveguide-with-ground-calculator.htm
JLCPCB JLC7628 4-Layer 1.2mm Impedance Control Stackup (Er = 4.6)
Ground is on layer 2.
Prepreg thickness: 0.2mm
12 mil track with 4 mil gap = 49.5 Ohms
CP2102N (USB-to-Serial Converter)
U4
VDD
VIO
RSTB
VREGIN
VBUS
D+
D-
GND
TXD
RXD
DCD
DTR
DSR
RTS
CTS
RI/CLK
SUSPEND
SUSPENDB
GPIO0/TXT
GPIO1/RXT
GPIO3/WAKEUP
GPIO2/RS485
CP2102N
C21
C22
C23
C24
0.1UF-0402T-6.3V-10%-X7R
4.7UF-0402-6.3V-20%-X5R
R5
22KOHM-0402T-1/16W-1%
R6
33KOHM-0402T-1/16W-1%
R7
2.2KOHM-0402T-1/16W-1%
U0TXD
U0RXD
Auto-Reset
Boot Mode Configuration
Pin | Default | Boot | Download
GPIO0 | 1 | 1 | 0
U0TXD | 1 | 1 | x
GPIO2 | 0 | x | 0
GPIO4 | 0 | x | x
MTDO | 1 | x | x
GPIO5 | 1 | 1 | x
If U0TXD, GPIO2, GPIO5 are floating, GPIO0 determines boot mode
Q1A
Q1B
MBT3904DW1T1
R8
R9
10KOHM-0402T-1/16W-1%
DTR
RTS
G0-BOOT
If DTR is LOW, toggling RTS from HIGH to LOW resets to run mode.
If RTS is HIGH, toggling DTR from LOW to HIGH resets to bootloader.
USB
J1
USB-C
VBUS
D+
D-
CC1
CC2
SHLD
GND
R10
R11
5.1k
smôl
open hardware
SPARK X
Special Instructions
Released under the Creative Commons Attribution Share-Alike 4.0 License
https://creativecommons.org/licenses/by-sa/4.0/
TITLE: ${PROJECTNAME}
Design by: >DESIGNER
Paul Clark
REV: >REV
Date: ${CURRENT_DATE}
Sheet: ${#}/${##}
Size: User
KiCad E.D.A. eeschema 7.0.6
Id: 1/1